14.4, 14.5 Gradients, part 1

If $z = f(x,y)$, the gradient of f is

$$\nabla f(x,y) = f_x(x,y)\,\vec{i} + f_y(x,y)\,\vec{j}$$

also denoted grad $f(x,y)$.

The gradient is the total derivative.

→ Do problems 1 and 2.

If $z = f(x,y)$ and $\vec{u}$ is a unit vector, the directional derivative of f at (a,b) in the direction $\vec{u}$ is

$$f_{\vec{u}}(a,b) = \nabla f(a,b) \cdot \vec{u}.$$

This is the instantaneous rate of change of f at (a,b) in the $\vec{u}$ direction.

→ Do problem 3.

Note

$$f_{\vec{u}}(a,b) = \nabla f(a,b) \cdot \vec{u} = \|\nabla f(a,b)\| \cos(\theta)$$

From this we get:

(1) ∇f points in the direction of greatest increase, i.e., it points toward the steepest upward slope.
(2) ∇f is orthogonal to contour lines.
(3) $f_{\vec{u}}(a,b) = 0$ when $\vec{u}$ is tangent to contour lines.

→ Do problems 4–6.